ChatGPT said:

Python/Django Assignment – AI-Assisted Box Selection System

Your assignment is to build a small **Django-based ecommerce warehouse management system** that automatically recommends the most suitable shipping box for customer orders.

I'll explain the complete requirements, simple business logic, implementation steps, and final expected output in easy English.

1. What is this project?

Imagine an ecommerce company sells different products.

For example:

- Mobile phone
- Laptop
- Shoes
- Books

When a customer places an order, the warehouse employee needs to decide:

Which shipping box is suitable for packing these products?

The system should automatically recommend a box based on:

1. Product dimensions (Length, Width, Height).
2. Product weight.
3. Box internal dimensions.
4. Box maximum weight capacity.
5. Box cost.

The main objective is to select the **smallest suitable box with the lowest cost**.







2. Understand the project requirements

You need to develop these five main functionalities.

Module	Requirement
Product Management	Add and manage products
Box Management	Add and manage shipping boxes
Order Management	Create customer orders
Box Recommendation	Automatically select a suitable box

Result Display	Show recommended box and cost

A. Product Management

Store product information.

Example:

Product	Length	Width	Height	Weight
Mobile	15 cm	8 cm	2 cm	0.2 kg
Laptop	35 cm	25 cm	3 cm	2 kg
Shoes	30 cm	20 cm	12 cm	1 kg

B. Box Management

Store available shipping box information.

Box	Length	Width	Height	Max Weight	Cost
Small	20	15	10	2 kg	₹10
Medium	40	30	20	5 kg	₹20
Large	50	40	30	10 kg	₹30

C. Order Management

The warehouse employee should be able to:

- Create a new order.
- Select one or more products.
- Enter product quantities.
- Submit the order.

D. Box Recommendation (Most Important)

This is the main business logic of your assignment.

Your system should:

1. Receive the selected products.
2. Calculate the total product weight.
3. Check whether the box can accommodate the products.
4. Check whether total weight is within the box capacity.
5. Compare all suitable boxes.
6. Recommend the most suitable box based on minimum cost and size.

Example:

Suppose a customer orders one mobile phone.

- Mobile dimensions: 15 × 8 × 2 cm.
- Mobile weight: 0.2 kg.

The system checks:

- Small box: Suitable.
- Medium box: Suitable.
- Large box: Suitable.

Recommended

Small

Box

–

₹10.

Why? Because it is the smallest suitable and cheapest box.

E. Result Display

After submitting an order, display:

Box Recommendation Result

Small Box Suitable



Details	Result
Product	Mobile
Total weight	0.2 kg
Box capacity	2 kg
Box cost	₹10
Recommendation	Small

If no box is suitable, display:

"No suitable box available for this order."

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3. Step-by-step implementation roadmap

Follow these steps in order.

1. **Step 1: Create Django Project**
   - Create a Django project.
   - Create a Django application named `box_selector`.
   - Configure the application.
   - Connect the database (SQLite is sufficient).
2. **Step 2: Create Database Models**

   Plan three main models:

   - Product
   - Box
   - Order

   Store dimensions, weight, capacity, quantity and cost.
3. **Step 3: Add Product Data**

   Create products through Django Admin.

   Example: Mobile, Laptop, Shoes.

4 **Step 4: Add Box Data**

Create Small, Medium and Large boxes through Django Admin.

5 **Step 5: Create Order Page**

Create a page where users can select products and quantities.

6 **Step 6: Implement Box Selection Logic**

This is the most important step.

- Calculate total weight.
- Check dimensions.
- Check weight capacity.
- Find suitable boxes.
- Select the lowest-cost suitable box.

7 **Step 7: Display Recommendation**

Show the recommended box, total weight, cost and reason for selection.

8 **Step 8: Test the Project**

Test different products, quantities and box sizes.

9 **Step 9: Prepare Submission Files**

Create README.md, AI_USAGE.md, test cases and TEST_OUTPUT.md.

10 **Step 10: Upload to GitHub**

Push your completed Django project to GitHub and submit the repository link.

4. Important business logic you must understand

There are two important calculations.

1. Total weight

For example:

- Mobile = 0.2 kg
- Laptop = 2 kg
- Quantity = 1 each

Total weight = 0.2 + 2 = 2.2 kg.

2. Box dimension checking

A product must fit inside the box.

For a single product, check whether its dimensions fit inside the box dimensions.

For multiple products, simply checking individual dimensions is not enough. You also need to consider how the products can be arranged together inside the box.

For your first implementation, you can use a simple approach:

- Check whether each product fits.
- Check total weight.
- If the order contains multiple products, use a clearly documented simplified packing rule or implement a proper packing algorithm.

Also, test different product orientations (rotating length, width and height).

5. Test cases required

Your project should include at least these test scenarios.

Test	Scenario	Expected result
1	Small product	Small box
2	Medium product	Medium box
3	Large product	Large box
4	Product exceeds box weight	Reject that box
5	Product exceeds all dimensions	No suitable box
6	Multiple products	Suitable box if packing is possible
7	Empty order	Show validation message
8	Multiple suitable boxes	Choose lowest-cost suitable box

6. Required submission folder structure

Your GitHub repository should look approximately like this:

ai-box-selection/

- config/
- box_selector/
- templates/
- tests/
- README.md
- AI_USAGE.md
- chat_transcript.pdf
- TEST_OUTPUT.md
- manage.py
- requirements.txt

7. What should you learn from this assignment?

The interviewer wants to evaluate these skills:

Skill	What they evaluate
Python	Logical thinking
Django	Project structure and models
Database	Storing and retrieving data
Business Logic	Box selection algorithm
Testing	Handling different scenarios
AI Usage	Your ability to use and verify AI output
GitHub	Project management and documentation

Important: Two special submission requirements

- `chat_transcript.pdf` : Export your actual ChatGPT conversation yourself. Do not ask AI to create a fake transcript.
- `What did you learn?` : Write this in your own words based on what you actually implemented and learned. Do not submit an AI-generated personal experience.

8. My recommendation: Complete the project in this order

Since your Django project setup has already started, don't start again from scratch.

Focus on these priorities:

1. Complete Product and Box database models.
2. Add sample products and boxes.
3. Understand and implement box selection logic.
4. Build the order creation page.
5. Display the recommendation.
6. Test all scenarios.
7. Prepare documentation and GitHub submission.

Most important

The box selection algorithm is the heart of this assignment.

Even

if

your

website

design

is

simple,

your

business

logic

must

work

correctly.

You can build this project with Python, Django, SQLite, HTML, CSS and Bootstrap. You don't need Machine Learning or a complicated AI model. The AI-assisted part means you can use AI development tools, but you must understand and verify your implementation.

Next, I can guide you through Step 1 onwards using your existing Django project, explaining each step in simple English and Marathi, without giving you code unless you ask for it.